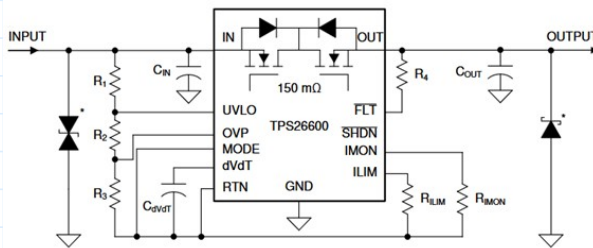


Detailed Design Procedure TPS2660



Design parameters:

$$V_{in} = 12V$$

$$V_{ov} = 12.6V$$

$$V_{uv} = 11.4V$$

$$R_L(su) = 12\Omega$$

$$I_{lim} = 2A$$

$$C_{out} = 2200\mu F$$

Undervoltage Lockout and Overvoltage Protection

From datasheet $V_{ovpr} = 1.19V$ and $V_{uvlor} = 1.19V$.

$$V_{ovpr} := 1.19 V$$

$$V_{uvlor} := 1.19 V$$

Overvoltage protection ($Ov = 12.6V$) and Undervoltage protection ($Uv = 11.4V$).
Values are from power supply (AKM45US12).

$$Ov := 12.6 V$$

$$Uv := 11.4 V$$

Resistor values calculated starting with $R3 = 30.1 k\Omega$ and equations from datasheet.
 $X = R1 + R2 + R3$.

$$R3 := 30.1 k\Omega$$

$$X := \frac{R3}{V_{ovpr}} \cdot Ov = 318.706 k\Omega$$

$$R2 := \frac{V_{uvlor}}{Uv} \cdot X - R3 = 3.168 k\Omega$$

$$R1 := X - R2 - R3 = 285.437 k\Omega$$

Looking for real resistor values (case code 0805 in) available on Mouser Electronics:

$$R1 := 287 k\Omega$$

$$R2 := 3.16 k\Omega$$

$$R3 := 30.1 k\Omega$$

Calculating new Voltage values:

$$V_{ov} := V_{ovpr} \cdot \frac{(R1 + R2 + R3)}{R3} = 12.661 V$$

$$V_{uv} := V_{uvlor} \cdot \frac{(R1 + R2 + R3)}{R2 + R3} = 11.458 V$$

Active Current Limiting

Proposed System max. current (2A) shall be the limit. The equation comes from datasheet:

$$I_{ol} := 2 A$$

$$R_{ilim} := \frac{12 kV}{I_{ol}} = 6 k\Omega$$

Using a real resistor value $R_{ilim} = 6 \text{ k}\Omega$:

$$R_{ilim} := 6 \text{ k}\Omega$$

$$I_{lim} := \frac{12 \text{ kV}}{R_{ilim}} = 2 \text{ A}$$

Input noise suppress

C_{in} has a typical capacitance value in range of $0.1 \text{ }\mu\text{F}$ to $1 \text{ }\mu\text{F}$.

$$C_{in} := 1 \text{ }\mu\text{F}$$

Current Monitoring Resistor

For $I_{lim}=2\text{A}$, and considering the operating voltage range of ADC from 0V to 2.5V :

$$V_{IMONmax} := 2.5 \text{ V}$$

$$R_{IMON} := \frac{V_{IMONmax}}{I_{lim} \cdot (75 \cdot 10^{-6})} = 16.667 \text{ k}\Omega \quad V := I_{lim} \cdot (75 \cdot 10^{-6}) \cdot 16.5 \text{ k}\Omega = 2.475 \text{ V}$$

Real value for R_{IMON} :

$$R_{IMON} := 16.5 \text{ k}\Omega$$

Output Voltage Ramp Time

For an inrush current $I_{inrush}=1\text{A}$ (estimated) shall be the t_{dVdT} calculated:

$$I_{inrush} := 1 \text{ A}$$

$$C_{out} := 2200 \text{ }\mu\text{F}$$

$$V_{in} := 12 \text{ V}$$

$$t_{dVdT} := C_{out} \cdot \frac{V_{in}}{I_{inrush}} = 65.34 \text{ ms}$$

For a given start-up time, C_{dVdT} capacitance value is calculated:

$$C_{dVdT} := \frac{t_{dVdT} \cdot 1 \text{ A}}{(8 \cdot 10^3) \cdot V_{in}}$$

$$C_{dVdT} = 275 \text{ nF}$$

The pull-up resistor (R_4 or R_{flt}) for the open-drain fault output shall have a value in the range of $10 \text{ k}\Omega$ to $100 \text{ k}\Omega$. The current sink by pin FLT must not exceed 10 mA .

$$R_{flt} := 100 \text{ k}\Omega$$

$$I_{flt} := \frac{12 \text{ V}}{R_{flt}} = 0.297 \text{ mA}$$